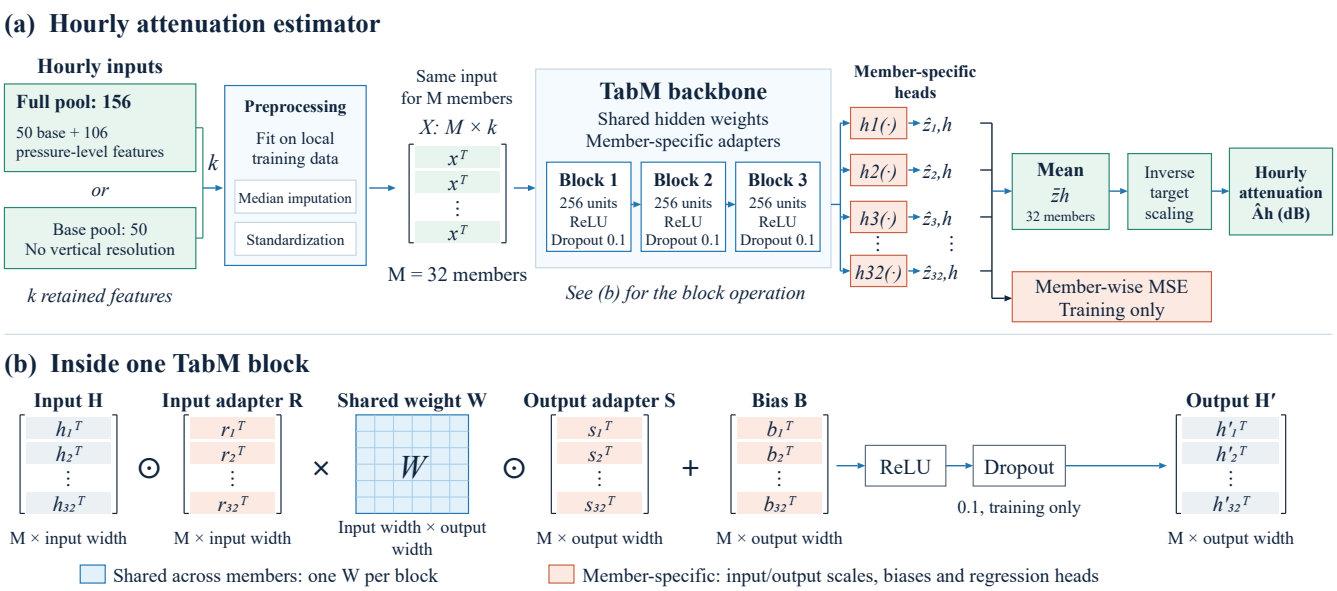

(a) Hourly attenuation estimator
Hourly inputs
Full pool: 156
50 base + 106 pressure-level features
or
Base pool: 50
No vertical resolution
k retained features
k
Preprocessing
Fit on local training data
Median imputation
Standardization
Same input for M members
X: M × k
M = 32 members
TabM backbone
Shared hidden weights
Member-specific adapters
Block 1
256 units
ReLU
Dropout 0.1
Block 2
256 units
ReLU
Dropout 0.1
Block 3
256 units
ReLU
Dropout 0.1
See (b) for the block operation
Member-specific heads
h1(·)
h2(·)
h3(·)
h32(·)
Mean
z̄h
32 members
Inverse target scaling
Hourly attenuation Âh (dB)
Member-wise MSE
Training only
(b) Inside one TabM block
Input H
M × input width
Input adapter R
M × input width
Shared weight W
Input width × output width
Output adapter S
M × output width
Bias B
M × output width
ReLU
Dropout
0.1, training only
Output H′
M × output width
Shared across members: one W per block
Member-specific: input/output scales, biases and regression heads